

Inhalational anesthetics: desflurane and sevoflurane.

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This article reviews the physico-chemical properties and performance characteristics of the two new potent inhaled anesthetics, desflurane and sevoflurane. Both drugs provide a greater degree of control of anesthetic depth and a more rapid immediate recovery from anesthesia than is currently available with other inhaled agents because of their decreased solubility. Desflurane is currently in widespread clinical use in the United States and parts of Europe. Compared with sevoflurane, it has the additional advantage of being extremely resistant to degradation and biotransformation. However, its pungent odor and tendency to irritate the respiratory tract make it unsuitable for inhalational inductions, and it has been linked to CO production in CO₂ absorbents. The sympathetic nervous system activation that occurs with desflurane limits its use in patients with cardiac disease. Otherwise, its hemodynamic and physiologic effects are similar to those seen with isoflurane. Studies of the economics of using desflurane are mixed, although it may offer the advantage of shorter postoperative recovery time. Sevoflurane is currently in widespread clinical use in Japan and parts of South America. The FDA Advisory Panel has recently recommended approval of sevoflurane in the United States, and we can expect the drug to be clinically available in the United States in the second quarter of 1995. Compared with desflurane, sevoflurane has the additional advantage of being nonirritating to the airway; inhalational induction of anesthesia with sevoflurane is achieved rapidly and easily. The instability of sevoflurane with CO₂ absorbents and its in vivo biotransformation produce potentially toxic byproducts. These byproducts, including Compound A and fluoride, have been extensively studied, and although the possibility for iatrogenic sequelae from sevoflurane exists, the likelihood of long-term toxicity appears quite low. Phase IV studies are indicated to determine the safety of administering sevoflurane (1) to renally impaired patients and (2) to any patient with fresh gas flows less than 2 L/min. Sevoflurane is otherwise very well tolerated and appears to offer the advantage of rapid and smooth induction and emergence from general anesthesia.

Desflurane. A review of its pharmacodynamic and pharmacokinetic properties and its efficacy in general anaesthesia.

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Desflurane is a halogenated ether inhalation general anaesthetic agent with low solubility in blood and body tissues, and approximately one-fifth the potency of isoflurane. The pharmacodynamic properties of desflurane generally resemble those of isoflurane; thus, it produces dose-dependent depression of the central nervous and cardiorespiratory systems, and tetanic fade at the neuromuscular junction. The alveolar equilibration of desflurane is rapid (90% complete at 30 minutes compared with 73% for isoflurane). Both desflurane and isoflurane are distributed to various tissues to a similar extent. Desflurane is resistant to chemical degradation and undergoes negligible metabolism (approximately equal to 10% of that seen with isoflurane). Desflurane 'wash-out' is approximately equal to 2 to 2.5 times faster than that of isoflurane in the first 2 hours after discontinuation of anaesthesia. The low solubility of desflurane facilitates a rapid induction of anaesthesia and precise control of the depth of anaesthesia (during maintenance). Results from a few clinical studies indicate that emergence from desflurane is significantly earlier (by approximately equal to 2 to 6 minutes) than that from propofol anaesthesia, whereas other studies do not concur. In comparison with isoflurane, emergence from desflurane anaesthesia is significantly earlier (by 5 minutes) after ambulatory and approximately equal to 50% earlier (also significant) after nonambulatory surgical procedures. Limited comparative studies with halothane or sevoflurane also suggest an earlier time of emergence from desflurane anaesthesia.

Comparative studies of desflurane and propofol, and other inhalation agents, indicate that the times to toleration of oral fluids, sitting and discharge from recovery room are similar, regardless of the general anaesthetic agent administered. However, some limited data in elderly patients (aged > 65 years) suggest that this patient group spends a significantly shorter time in the postanaesthesia care unit after desflurane than after isoflurane anaesthesia. Differences, if any, in the recovery of cognitive and psychomotor functions after desflurane or propofol anaesthesia remain unclear. However, in comparison with isoflurane anaesthesia, recovery of these functions (up to 45 minutes post-operatively) occurs earlier after desflurane. Significantly fewer patients are subjectively impaired (i.e. drowsy, clumsy, fatigued or confused) upon recovery from desflurane than from isoflurane anaesthesia. Likewise, significantly fewer adult patients are delirious when recovering from desflurane than from isoflurane anaesthesia, though in paediatric patients delirium is more likely when recovering from desflurane than from halothane anaesthesia. Haemodynamic stability during coronary artery surgery is as well maintained with desflurane as with isoflurane, and the drug does not worsen the adverse postoperative outcomes.(ABSTRACT TRUNCATED AT 400 WORDS)

Other new and potentially useful inhalational anesthetics.

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Sevoflurane and desflurane are volatile inhaled anesthetics that are currently being investigated as possible improvements for the anesthetic management of human patients. Information to date suggests these agents have several advantages over existing clinical agents. For example, the blood/gas partition coefficient for both agents is lower than that of other halogenated anesthetics. Consistent with this physical characteristic is a more rapid induction of and emergence from anesthesia. Both cause a dose-related depression of cardiopulmonary function, which is comparable to isoflurane. Results of studies to date favor desflurane over sevoflurane because it is less soluble in blood, is stable in soda lime, is biodegraded the least of any volatile anesthetic, and is not toxic.

New halogenated agents: should I change my practice?

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Sevoflurane and Desflurane are relatively new halogenated agents which make induction and control of depth of anaesthesia easier, recovery rapid and of good quality and they have less side-effects and toxicity. In children sevoflurane could replace halothane because it provides smooth and rapid induction with less cardiovascular depression and arrhythmias. Desflurane is not used because of its pungent odour. In adults sevoflurane could be preferred to desflurane because it allows rapid induction and laryngeal mask insertion or tracheal intubation without myorelaxants, a similar time of recovery, no clinical evidence for renal and hepatic toxicity, no more costs for anaesthesia for a lower MAC.

Desflurane and sevoflurane. New volatile anesthetic agents.

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Desflurane and sevoflurane, recently licensed for use in humans, have kinetics that result in rapid induction and easy maintenance of a stable level of anesthesia. Recovery is also rapid. Cardiopulmonary effects are similar to those of isoflurane. In humans, desflurane can cause airway irritation and sympathetic stimulation, but these side effects have not caused problems in animal trials. Metabolites of sevoflurane and breakdown products from its reaction with carbon dioxide absorbents theoretically could result in hepatic and renal damage, but such toxicity has not occurred despite extensive medical use. Clinical trials in animals are now in progress.

A prospective randomized study comparing recovery following anesthesia with a combination of intravenous dexmedetomidine and desflurane or sevoflurane in spinal surgeries.

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Desflurane and sevoflurane are inhalational anesthetics which provide stable intraoperative hemodynamics and rapid emergence from anesthesia. Dexmedetomidine is an α_2 -agonist with sedative and hypnotic effects. We compared recovery following anesthesia with a combination of a continuous intravenous infusion of dexmedetomidine and desflurane or sevoflurane in cases of spine surgeries because no such data are available from India. It was a single-blind, prospective, randomized study. After institutional ethics committee approval, patients were randomly allocated to one of the two groups of fifty patients each. Group D received desflurane and Group S received sevoflurane, along with dexmedetomidine 0.5 $\mu\text{g/kg/h}$ IV infusion for maintenance of anesthesia. Extubation time (ET) in Group D was shorter by 4.2 min than in Group S (10.1 ± 2.2 and 14.2 ± 1.3 ; $P = 0.004$). Postoperative recovery, postoperative analgesic, and antiemetic requirement between the groups were comparable. The mean dial setting required to maintain the minimum alveolar concentration of 1 intraoperatively was 3.1 for desflurane and 0.7 for sevoflurane.

A comparison of postoperative respiratory complications associated with the use of desflurane and sevoflurane: a single-centre cohort study.

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Sevoflurane and desflurane are the most commonly used volatile anaesthetics for maintenance of anaesthesia. In this study, we aimed to evaluate the relationship between choice of volatile anaesthetic and early postoperative respiratory complications, and to address a critical knowledge gap in safety outcomes between these two commonly used agents. We performed a retrospective analysis of adult (non-cardiac surgery) patients who received sevoflurane or desflurane for the maintenance of general anaesthesia at our institution between 2005 and 2018. We evaluated the association between desflurane exposure (when compared with sevoflurane) and the primary outcome of postoperative respiratory complications, defined by early post-extubation desaturation ($\text{Sp O}_2 < 95\%$); elderly (age > 65 years); and high risk of respiratory complications as well as the primary outcome at 24 h. Desflurane was used for 23,830 patients and sevoflurane for 84,608 patients. Patients exposed to desflurane did not demonstrate a reduced risk of postoperative respiratory complications when compared with sevoflurane (adjusted odds ratio 0.99, 95%CI 0.94-1.04, $p = 0.598$). These findings were consistent across all sub-groups of high-risk patients and in the propensity score matched cohort. In summary,

desflurane use was not associated with reduced postoperative respiratory complications when compared with sevoflurane. In the context of environmental and cost concerns with volatile anaesthetic agents, our study provides important data to support organisational decisions regarding the use of desflurane.

Gyosaponin ameliorates sevoflurane anesthesia-induced cognitive dysfunction and neuronal apoptosis in rats through modulation of the PI3K/AKT/mTOR pathway.

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Sevoflurane (Sev) is an inhalational anesthetic for surgical procedures where it can trigger cognitive dysfunction and neuronal apoptosis. Gyosaponin (GpS) was studied for its effects on brain morphology and cognitive behaviors in Sev-anesthetized rats. Male Sprague-Dawley rats were induced by 3 % Sev anesthesia, and 25 mg/kg and 100 mg/kg GpS were injected into the rats by tail vein. The in vitro model of Sev anesthesia was constructed by treating primary rat hippocampal neurons with 4.1 % Sev in the presence of GpS (5, 10, and 20 μ M). The neuroprotective effects of GpS against Sev-induced cognitive deficits in rats were evaluated using the open field and Morris water maze tests. The apoptosis of hippocampal neurons was observed using HE staining and TUNEL assay. Apoptosis-related proteins and proteins related to the PI3K/Akt/mTOR pathway were determined via Western blot. Also, pro-inflammatory factors were measured via ELISA. GpS diminished the Sev-triggered apoptosis in neurons and Cleaved caspase-3, BAX, TNF- α , IL-6, lessened oxidative stress damage, and stimulated the PI3K/Akt/mTOR pathway. GpS therapy markedly enhanced learning and memory abilities in rats suffering from Sev-related cognitive impairments. GpS ameliorates Sev-induced neurotoxicity and cognitive dysfunction by modulating the PI3K/Akt/mTOR pathway and alleviating neuronal apoptosis and oxidative stress.

Effect of vasoconstriction by dexmedetomidine in the mandible of rat.

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In recent years, dexmedetomidine (DEX) has been proposed as a useful vasoconstrictor for local anesthesia because it is less effective in circulation than clonidine of antihypertensive drugs. In addition, DEX is expected to act as a vasoconstrictor during local anesthesia. However, histomorphometric studies demonstrating that DEX exerts vasoconstrictive effects are lacking. This study aimed to clarify whether DEX exerts a histomorphologically vasoconstrictive effect on blood vessels in the mandible of rats. A total of 12 male Wistar rats were used. General anesthesia was induced and maintained using sevoflurane. Normal saline (0.2 ml) was injected on the left side of the jaw (DEX (-) effect site) and 0.2 ml normal saline containing 12.5 μ g/ml DEX was injected on the right side of the jaw (DEX (+) effect site). The puncture point was located on the mesial side of the first molar, 1 mm away from the gingival sulcus. Following decalcification, the specimens were paraffinized and sagittally sliced into 20 μ m-thick sections, followed by staining with anti- α smooth muscle actin antibody. The intravascular lumen area was measured in the oral mucosa, periodontal ligament, mandibular bone above the root apex, mandibular bone below the root apex, and dental pulp. The unpaired t-test was used for statistical analysis, and a P value < 0.05 was considered statistically significant. Compared to the DEX (-) effect site, the intravascular lumen area in the oral mucosa and periodontal ligament of the DEX (+) effect site was significantly decreased. No significant difference was observed in the intravascular lumen area between

the DEX (-) and DEX (-) effect sites in the mandibular bone above and below the root apex and dental pulp. A direct vasoconstrictive effect of DEX was not observed in the intravascular lumen of the mandibular bone above and below the root apex and dental pulp; however, it was observed in the oral mucosa and periodontal ligament.

Desflurane and sevoflurane use during low- and minimal-flow anesthesia at fixed vaporizer settings.

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The pharmacokinetics for sevoflurane and desflurane makes them suitable for low-flow anesthesia. The aim of the present study was to assess the use of desflurane and sevoflurane at constant vaporizer settings and fixed low fresh gas flows. One hundred ASA 1-2 patients undergoing elective laparoscopic surgery were randomized into 4 groups (25 patients each): a fixed fresh gas flow 1.0 or 0.5 L/min with desflurane (D1.0 and D0.5) or sevoflurane (S1.0 and S0.5) throughout anesthesia. A fixed vaporizer setting, sevoflurane 6% and desflurane 18% was used during wash-in. Time to reach 1 and 1.5 minimum alveolar concentration (MAC), emergence and gas consumption from start to end of surgery was studied. Time to reach 1 MAC age adjusted desflurane or sevoflurane was D0.5 8.5 ± 1.7 , D1.0 3.7 ± 0.7 , S0.5 15.2 ± 2.4 and S1.0 6.2 ± 1.3 minutes, respectively ($P < 0.001$), and times to increase from 1 to 1.5 MAC differed also significantly. Desflurane anesthesia was associated to significantly shorter time to extubation 6.7 ± 2.3 vs. 10 ± 2.3 minutes for sevoflurane ($P < 0.001$). The amount of agent consumed g/min. was significantly reduced for both 0.5 L/min groups: 30% less for desflurane and 19% for sevoflurane. We found an almost twice as rapid wash-in with desflurane and expectedly faster emergence. Gas consumption was lower at 0.5 L/min than it was at 1 L/min for both gases studied however most pronounced for desflurane. Desflurane has clear advantages for minimal fresh gas flow anesthesia.
